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Zimmamless — V3 Consolidated System Requirements

Version: 3.0 **Date:** July 2026 **Jurisdiction of operation:** Hashemite Kingdom of Jordan

Status: Competition-planning authority document **Supersedes:**

Zimmamless_Complete_System_Requirements_EN1.pdf (V2.0),

Zimmamless_Logical_Class_Diagram_Report.md,

Zimmamless_Logical_Class_Diagram.mermaid,

zimmamless_workflow_diagram.html

Document Control

Authority

Where any previous Zimmamless document conflicts with this document, **this document governs**. The V2.0 PDF, the earlier logical class diagram, and the earlier workflow diagram are retained for historical reference only and must not be used as an implementation source.

Legal disclaimer and the **LEGAL_TBD_POST_COMPETITION** convention

Zimmamless operates in a domain touching financial regulation, assignment of receivables, data protection, and contract enforceability. **No legal, regulatory, licensing, contractual, enforceability, liability, privacy, data-residency, assignment, perfection, or formal-notification question is settled by this document.**

All such questions are marked inline with the tag:

`LEGAL_TBD_POST_COMPETITION`

and are additionally collected in **Appendix A — Legal TBD Register**.

These items:

- will be resolved after the competition with qualified legal and financial specialists;
- **must not block** product definition, architecture, database design, API design, UX, or implementation planning;
- are handled in the technical system through **configurable policies, provider adapters, document templates, and enumerations**, so that final legal rules can be applied later without redesigning the core platform.

Every `LEGAL_TBD_POST_COMPETITION` item in this document is accompanied by the **current technical assumption** the platform implements in the interim.

Reading conventions

Marker	Meaning
MUST	Mandatory requirement for the V3 release.
SHOULD	Strongly recommended; deviation requires a recorded decision.
MAY	Optional or future-phase.
CONFIGURABLE	Value or behaviour is an administrator-managed platform policy, not a hard-coded constant.
LEGAL_TBD_POST_COMPETITION	Legal question deferred; technical assumption stated alongside.
ASSUMPTION	A gap filled by the authoring team, subject to owner override in review.

Requirement identifiers

Requirements are identified as ZM- <MODULE> - <NNN>, e.g. ZM- OFR- 014. Module codes are listed in §4.2.

1. Executive Summary

Zimmamless is a **digital receivables marketplace** operating in Jordan. It connects **suppliers** who hold unpaid, deferred commercial invoices with **licensed banks** willing to advance funds against those invoices.

The journey in one paragraph: a supplier registers using its national establishment number; the platform enriches the supplier profile automatically from government registries and reaches a decision within a 24-business-hour service level. The supplier then uploads an electronic invoice, identifies the buyer (who is resolved against the companies registry but is never a platform user), and states a private minimum acceptable net amount. The platform runs automated verification, duplicate detection, fraud screening, and a versioned Trust Score, then publishes the invoice as a **confidential, time-boxed marketplace listing** visible only to banks whose policy filters match. Banks submit competing offers, invisible to one another; only the supplier can compare them. The supplier accepts one offer at any time while the listing is open — acceptance is atomic and permanently locks the invoice. A contract is generated from a template, both parties sign electronically, the bank funds, and a **cross-party OTP** confirms funding participation from both sides. The platform then settles: gross funding in, platform commission and any unpaid listing fee deducted, net payout to the supplier. After funding, Zimmamless **continues to track the receivable** — maturity, partial payment, full payment, overdue, recourse, dispute, and closure — even though the buyer's payment to the bank happens entirely outside the platform.

1.1 What Zimmamless is

- An **electronic marketplace** and workflow platform.

- A **verification and data-enrichment** layer over Jordanian government registries.
- A **decision-support** provider producing transparent, versioned, explainable risk indicators.
- A **settlement orchestrator** for the single bank-to-supplier money movement it touches.
- A **system of record** with an immutable audit trail across the full receivable lifecycle.

1.2 What Zimmamless is not

- **Not a lender.** The platform never advances its own funds.
- **Not a guarantor.** It does not guarantee buyer repayment or bank recovery.
- **Not an adjudicator.** It records disputes and recourse cases; it does not decide them.
- **Not a collector.** Buyer-to-bank payment happens directly, outside the platform.
- **Not an auction.** There is no live bidding, no visible competing prices, and no automatic winner.

1.3 The ten defining rules of V3

1. **Neutral transaction model.** The core entity is `ReceivableTransaction`; the legal shape is a per-offer `transactionType`.
 2. **Transparent money.** Every offer discloses a full deduction breakdown and a single net payout figure.
 3. **Private floor.** The supplier's `minimumAcceptableAmount` is never visible to banks.
 4. **Confidential offers.** No bank sees another bank's identity, amount, fees, or conditions.
 5. **Immediate acceptance.** The supplier may accept at any time while the listing is open; acceptance is atomic and irreversible.
 6. **No cheques.** Post-dated cheque handling is entirely out of scope in V3.
 7. **Cross-party funding confirmation.** OTP participation from both bank and supplier is mandatory before FUNDED.
 8. **Lifecycle continues after funding.** Monitoring, overdue, recourse, and dispute tracking are in scope.
 9. **Never punish missing government data.** Source downtime or unpublished fields reduce data confidence; they never count as adverse evidence.
 10. **Everything legal is configurable.** Templates, fee policies, penalty rules, recourse types, and retention rules are data, not code.
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2. Change Log — V2 to V3

2.1 Removed

Removed	Rationale
All post-dated cheque functionality — <code>PostDatedCheque</code> , <code>HandoverAppointment</code> , <code>ChequeHandover</code> , <code>HandoverReceipt</code> , cheque encryption/masking, in-person	Cheques are out of scope for V3. Any future cheque structure is LEGAL_TBD_POST_COMPETITION.

Removed	Rationale
handover meeting, REQUIRED_POST_DATED_CHEQUE condition type, cheque status enums	
Physical-presence requirement for any platform step	No step in V3 requires the parties to meet in person.
requestedFinancingAmount field	Replaced by <code>minimumAcceptableAmount</code> as the single supplier-side amount input.
Selection-after-deadline-only rule (from the class diagram)	Superseded: the supplier may accept while the listing is open.
Hard-coded purchase-only model (<code>ReceivablePurchaseContract</code> , <code>acceptedPurchaseAmount</code> as the sole amount)	Superseded by the neutral <code>ReceivableTransaction</code> with configurable <code>transactionType</code> .
Per-submission Buyer composition	Superseded: Buyer is a global, deduplicated platform entity.
Stripe named in the domain model	Replaced by a provider-neutral settlement adapter.

2.2 Retained from V2 (PDF)

Retained	Notes
Government-first data model — supplier gives minimum data, platform enriches	Core to the product.
Self-declared data never overwrites government-source data	Retained verbatim.
24-business-hour onboarding SLA with clock pause/resume	Retained; business calendar now explicitly configurable.
Application states including <code>INFORMATION_REQUIRED</code> , <code>GOVERNMENT_SERVICE_UNAVAILABLE</code> , <code>APPROVED_CONDITIONAL</code>	Retained.
Hard-rejection rules	Retained and extended.
Detailed buyer search and identity resolution, <code>BUYER_MANUAL_REVIEW</code>	Retained at full rigour.
Buyer is never a platform user	Retained and reinforced.
Supplier invoice declarations	Retained and extended.
Automated invoice checks (completeness, identity, duplicate, logic, eligibility, integrity)	Retained.
Five risk indicators	Retained, now alongside a composite score.
Government downtime must not reduce the score	Retained as a first-class rule.

Retained	Notes
Post-funding monitoring, overdue, recourse, dispute, fraud modules	Retained — explicitly back in scope.
Immutable audit log; financial records never deleted	Retained.
2.3 New in V3	
New	Summary
ReceivableTransaction neutral core entity	Single lifecycle across all transaction types.
transactionType per offer	INVOICE_FINANCING, RECEIVABLE_PURCHASE, RECEIVABLE_ASSIGNMENT, OTHER.
recourseType as a first-class offer field	FULL_RECOURSE, LIMITED_RECOURSE, NON_RECOURSE, OTHER.
Full offer money breakdown	Six deduction components plus computed net payout.
minimumAcceptableAmount as a net floor	Compared against netSupplierPayout, not gross.
Partial funding	A bank may fund less than the full outstanding amount.
Listing fee	Incurred at listing activation, independent of funding outcome.
Cross-party OTP funding confirmation	Mandatory; replaces the removed cheque-handover OTP with a different purpose.
OVERDUE_UNCONFIRMED state	Distinguishes “bank has not reported” from “confirmed default”.
WithdrawalCase with structured reason codes and configurable penalty policy	Post-acceptance bank withdrawal handling.
Composite Trust Score (0–100) + risk band	Alongside the five component indicators.
Versioned scoring rules	Historical scores retain their calculation version.
Real ML pipeline	Training, inference, versioning, metrics, explainability, fallback.
OCR + QR extraction and cross-validation	Mandatory electronic invoice; QR decoded and compared.
Maker/approver separation for bank offers	Mandatory; self-approval prohibited.
Multi-organization user membership with active-context switching	Many-to-many.

New	Summary
Double-entry ledger for platform-controlled financial legs	Reversals via compensating entries only.
Bilingual English/Arabic with full RTL	First-class requirement.
Demo time machine	Non-production maturity simulation.
LEGAL_TBD_POST_COMPETITION register	Formal deferral mechanism.

2.4 Revised

Area	V2	V3
Core transaction	Invoice financing with recourse (PDF) / outright purchase (class diagram) — contradictory	Neutral ReceivableTransaction with per-offer transactionType
Offer acceptance timing	Immediate (PDF) vs. after deadline (class diagram) — contradictory	Immediate, while listing open
Supplier amount input	requestedFinancingAmount	minimumAcceptableAmount (net floor, private)
Buyer entity	Per-submission composition	Global deduplicated entity + SupplierBuyerRelationship
Money flow	Bank pays supplier directly; platform fee unclear	Bank → configurable settlement adapter → commission + listing fee deducted → net payout
Commission recognition	Unspecified	Finalized only on SUPPLIER_PAYOUT_COMPLETED
Payment provider	Stripe adapter named	Provider-neutral SettlementProvider adapter
Risk output	Five indicators	Composite score + five indicators + reason codes + version
Deadlines	Undefined	Admin-configurable defaults: 24h offers, 12h selection

3. Stakeholders and Actors

3.1 Actor summary

Actor	Platform account	Core responsibility
Supplier organization	Yes	Registers the business,

Actor	Platform account	Core responsibility
		uploads invoices, resolves buyers, sets a minimum acceptable amount, compares offers, accepts one, signs, confirms funding, responds to recourse.
Bank organization	Yes	Configures policy filters, reviews eligible listings, creates and approves offers, signs, funds, confirms funding via OTP, reports buyer payments, initiates recourse.
Platform administration	Yes	Verifies supplier applications, manages the marketplace, configures fees and policies, reviews exceptions, supports users.
Compliance function	Yes	Reviews fraud cases, mismatches, sanctions concerns, and blocked entities.
Buyer / debtor	No	Stored as a resolved company record; owes the invoice; receives an outbound notification; has no login, no offers, no decision authority.
Government registries	N/A (system)	CCD, ISTD, GAM/licensing — read-only data sources via adapters.
Settlement provider	N/A (system)	Executes the bank-to-supplier money movement via adapter.
Signature provider	N/A (system)	Executes and verifies electronic signatures via adapter.

3.2 The buyer's position — restated

ZM-BUY-001 The buyer **MUST NOT** have a platform account, login, portal, or read-only view. ZM-BUY-002 The buyer **MUST NOT** approve, reject, or influence any offer, contract, or funding decision. ZM-BUY-003 The buyer **MUST** be resolved to a verified registry identity before an invoice becomes eligible. ZM-BUY-004 The buyer **MUST** receive an outbound notification after the transaction is confirmed (§13). ZM-BUY-005 Buyer contact details **MUST** be stored against

the SupplierBuyerRelationship, not the global Buyer, because different suppliers legitimately hold different contacts at the same company.

LEGAL_TBD_POST_COMPETITION — **LT-01**. Whether the buyer notification constitutes formal notice of assignment, whether buyer acknowledgement is legally required, and the legal basis for the platform processing buyer contact data provided by a third party (the supplier). **Technical assumption:** the platform sends an operational notification, stores full delivery evidence, supports correction and invalidation of contact data, and restricts access by role.

3.3 Role catalogue

3.3.1 Bank roles

Role	Capabilities
Bank Admin	Manage bank users, roles, policy filters, integration settings, disbursement accounts.
Viewer / Analyst	View eligible listings and full permitted data; run analysis; cannot create or approve offers.
Offer Maker	Create, revise, and submit an offer for internal approval; withdraw before acceptance.
Offer Approver	Approve or reject an offer created by another user. MUST NOT approve an offer they created.
Operations	Mark funding sent, generate funding OTP, record buyer payments, attach evidence, initiate recourse.
Auditor	Read-only across the bank's own records and reports; no write capability anywhere.

ZM-ROL-001 Maker/approver separation **MUST** be enforced for every published offer in V3.
ZM-ROL-002 The system **MUST** reject any approval where approvedByUserId == createdByUserId.

3.3.2 Supplier roles

Role	Capabilities
Supplier Owner / Admin	Manage the organization, users, bank accounts, and consents; full visibility.
Authorized Signatory	Sign contracts on behalf of the supplier; must match or be authorized against registry signatories.
Invoice Uploader	Create submissions, resolve buyers, upload documents, set the minimum acceptable amount.
Viewer	Read-only.

ZM-R0L-003 A single user **MAY** hold multiple supplier roles simultaneously. ZM-R0L-004 Only an **Authorized Signatory** may sign; only an Owner/Admin or Uploader may accept an offer, per CONFIGURABLE policy. ASSUMPTION — default: offer acceptance requires Owner/Admin.

3.3.3 Platform roles

Role	Capabilities
Super Admin	Full system configuration, role management, policy versions, emergency controls.
Operations Admin	Marketplace operation, listings, relisting approvals, withdrawal cases, fee configuration.
Supplier Reviewer	Onboarding review, information requests, approval/rejection decisions.
Compliance Officer	Fraud cases, sanctions review, blacklisting, restriction decisions.
Support Agent	User assistance; read-mostly; no financial or decision authority.
Platform Auditor	Full read access to audit logs and reports; no write capability.

3.3.4 Multi-organization membership

ZM-R0L-005 A User **MAY** hold memberships in multiple organizations (many-to-many OrganizationMembership). ZM-R0L-006 The user **MUST** operate within exactly one **active organization context** at a time. ZM-R0L-007 All permissions, queries, and data access **MUST** be scoped to the active context; cross-context data leakage is a critical defect. ZM-R0L-008 Every audit entry **MUST** record both actorUserId and actorOrganizationId (the active context). —

4. System Scope and Module Map

4.1 V3 release scope

The competition release delivers the **complete internal platform**. External dependencies (government registries, settlement, signatures, messaging) run through **adapters** with dummy/sandbox implementations, but every internal workflow **MUST** genuinely function end to end.

In scope: authentication; roles and permissions; multi-org context; supplier onboarding with SLA; government verification workflow; buyer resolution; invoice submission; document storage; OCR; QR extraction; verification checks; duplicate detection; fraud review; Trust Score with real ML; bank onboarding with demo organizations and real user accounts; policy filters and eligibility; marketplace listings; confidential offers; maker/approver approval; supplier offer comparison and selection; atomic invoice locking; offer conditions; contract generation and electronic signature; funding settlement; cross-party OTP; listing fee; platform

commission; supplier payout tracking; buyer-to-bank payment status tracking; partial payment; overdue tracking; recourse; disputes; withdrawal cases; notifications with delivery evidence; administration; compliance; immutable audit logs; reports; bilingual EN/AR with RTL; demo time machine.

Out of scope for V3: post-dated cheques and any handover process; native mobile applications; buyer portal of any kind; real production government API integration; real regulated fund custody; production PKI; multi-currency; secondary market or invoice resale; automatic relisting.

4.2 Module codes

Code	Module
IAM	Identity, Access, and Organizations
SON	Supplier Onboarding
GOV	Government Verification
BUY	Buyer Directory and Resolution
INV	Invoice Submission and Management
DOC	Document Storage, OCR, and QR
VER	Verification and Duplicate Detection
RSK	Risk, Trust Score, and ML
MKT	Marketplace and Listings
OFR	Bank Offers and Conditions
SEL	Offer Selection and Locking
CON	Contracts and Signatures
FND	Funding, OTP, and Settlement
FEE	Fees, Commission, and Ledger
PMT	Post-Funding Payment Tracking
REC	Recourse, Disputes, Withdrawal
FRD	Fraud and Compliance
NOT	Notifications
AUD	Audit, Reporting, Administration
I18N	Internationalization
DEM0	Demo and Test Tooling

5. Supplier Onboarding

5.1 Principle

ZM-SON-001 The supplier **MUST** provide only the minimum data that cannot be obtained from a government source. ZM-SON-002 The platform **MUST** retrieve all available registry data automatically. ZM-SON-003 Government-sourced values **MUST NOT** be editable by any

user, including administrators. Corrections are made by re-querying the source. ZM-SON-004 Supplier-provided values **MUST** be stored as SELF_DECLARED with evidence and a verification status, and **MUST NOT** overwrite a government-sourced value for the same field.

5.2 Data entered by the supplier

Stage	Fields	Purpose
Account creation	Phone, email, password	Login, OTP, notifications
Business identification	National establishment number; profession licence number	Registry lookup keys
User identity	National ID of the account user	Match against authorized signatories
Disbursement	IBAN, bank name, account-holder name, ownership evidence	Ensure payout reaches the correct business account
Consents	Lookup and sharing authorization; terms; privacy; declarations	Enable verification and bank disclosure

ZM-SON-005 The platform **MUST NOT** ask the supplier to type: company name, legal type, status, registration date, registered address, capital, partners, authorized signatories, business purposes, or tax number. These are derived.

5.3 Data retrieved automatically

Source	Key	Retrieved
Companies Control Department (CCD)	National establishment number	Name, company number, type, status, registration and modification dates, address, registered contacts, capital, authorized signatories, purposes, partners, management, announcements
Income and Sales Tax Department (ISTD)	National establishment number	Tax number, registration status, registered name (subject to availability)
Greater Amman Municipality / licensing authority (GAM)	Establishment number + profession licence number	Licence status, activity, address, issue and expiry dates (subject to availability)

ZM-GOV-001 Every lookup **MUST** persist a raw GovernmentDataSnapshot (verbatim source payload) **and** a normalized result object. ZM-GOV-002 Every field **MUST** carry: value, source, retrievedAt, verificationStatus, evidenceRef, sourceReference. ZM-GOV-003 Registry results are explicitly **preliminary and possibly incomplete**. Blank fields are normal and **MUST NOT** be treated as adverse.

5.4 Verification sequence

1. Account created; phone and email verified.
2. Supplier enters establishment number and licence number, and grants lookup authorization.
3. Platform queries CCD; snapshot and retrieval timestamp persisted.
4. Company status checked; core name and registration identifiers matched.
5. ISTD queried; tax number retrieved; registered name matched.
6. GAM queried; licence status, activity, and address checked.
7. Account user compared against authorized signatories; if absent, authorization evidence requested.
8. IBAN and account ownership verified.
9. Automated screening completed, then human review.
10. Decision issued: approval, conditional approval, information request, or rejection.

5.5 Application states and the 24-business-hour SLA

State	Meaning	SLA clock
DRAFT	Not yet submitted	Not started
SUBMITTED	Application submitted	Starts
AUTOMATED_VERIFICATION	Registry checks running	Running
UNDER_REVIEW	Human reviewer assessing	Running
INFORMATION_REQUIRED	Awaiting supplier input	Paused
INFORMATION_RESUBMITTED	Supplier responded	Resumes
GOVERNMENT_SERVICE_UNAVAILABLE	Source unavailable	Paused
FINAL_REVIEW	Decision review	Running
APPROVED / APPROVED_CONDITIONAL / REJECTED	Decided	Stops

ZM-SON-006 Target: a decision within **24 business hours** after the file is complete and required government services are available. ZM-SON-007 Business calendar defaults (CONFIGURABLE): **Sunday–Thursday, 08:00–17:00, Asia/Amman**, with a configurable public-holiday calendar. ZM-SON-008 Every pause and resume **MUST** be recorded with timestamp, reason, and actor, so elapsed business time is fully reconstructible. ZM-SON-009 The supplier **MUST** see remaining SLA time and current state at all times.

5.6 Missing information handling

Case	System action	Request to supplier
Optional field absent	No effect on decision	None, or optional completion
Essential field absent	INFORMATION_REQUIRED	Provide value + evidence
Authorized signatory absent	Suspend activation	Recent signatory certificate

Case	System action	Request to supplier
Beneficial owner unavailable	Separate request when required	or authorization Ownership declaration + evidence
Licence not found	Review or correction request	Correct number or current copy
Bank account mismatch	Mandatory manual review	Ownership evidence or corrected account

ZM-SON-010 Government-service downtime **MUST NOT** reduce any reliability indicator and **MUST NOT** cause rejection.

5.7 Decision outcomes

Outcome	Conditions	Result
Approval	Active company; tax, licence, signatory, and bank account verified; no hard blocker	APPROVED → ACTIVE
Conditional approval	Non-material operational item outstanding	APPROVED_CONDITIONAL; account accessible, financing actions disabled until conditions cleared
Information request	Essential data missing or correctable mismatch	INFORMATION_REQUIRED
Rejection	Hard-rejection condition, fraud, or business cannot be established	REJECTED with a structured reasonCode

ZM-SON-011 Under APPROVED_CONDITIONAL, the supplier **MUST** be able to log in, view the platform, and complete outstanding items, but **MUST NOT** create invoice submissions or listings.

5.8 Hard-rejection rules

ZM-SON-012 The following **MUST** produce rejection:

- Company suspended, struck off, or not found.
- Company in liquidation or insolvency, subject to the approved legal and credit policy. LEGAL_TBD_POST_COMPETITION — **LT-02**. Treatment of liquidation. **Technical assumption:** configurable policy; default is manual review escalating to rejection.
- Profession licence suspended or cancelled, or activity prohibited.
- Forgery, manipulation, or impersonation of an authorized signatory.
- Refusal of essential consents, or inability to establish ownership of the bank account.
- Confirmed legal prohibition or mandatory sanctions-list match.

ZM-S0N-013 **Sole proprietorships and entities that cannot be verified** through CCD or a supported licensing authority are **not eligible** in V3. The platform **MUST** display a clear, non-pejorative ineligibility message and record the attempt.

5.9 Snapshot freshness and re-verification

ZM-GOV-004 A government snapshot is considered current for **90 days** (CONFIGURABLE). ZM-GOV-005 The platform **MUST** re-query when a new invoice is submitted and the most recent relevant snapshot is older than the freshness window. ZM-GOV-006 V3 **MUST NOT** run scheduled background re-verification sweeps. Re-verification is **activity-triggered** only. ZM-GOV-007 If a re-check reveals the supplier has become suspended or struck off, the platform **MUST**: suspend or restrict the account; block new invoice submissions; preserve all existing transactions and financial records unchanged; notify administration and the supplier.

5.10 Government adapter contract

ZM-GOV-008 Each source **MUST** be implemented behind a common adapter interface with dummy and production implementations, delivering:

- documented endpoints, request and response schemas, and status codes;
- **full, partial, and unavailable** response variants;
- deterministic seeded test data keyed by establishment number;
- simulated latency and configurable failure injection;
- authentication placeholders;
- retry, timeout, and circuit-breaker behaviour;
- raw snapshot persistence plus normalized output;
- a `sourceAvailability` signal distinguishing “*source said no*” from “*source did not answer*”.

ZM-GOV-009 Replacing a dummy adapter with a production adapter **MUST NOT** require any change to core domain logic.

LEGAL_TBD_POST_COMPETITION — **LT-03**. Permissible storage, sharing, reuse, and retention of government-sourced data, and whether supplier consent covers onward disclosure to banks. **Technical assumption**: consent recorded per purpose and version; snapshots retained; disclosure to banks gated by policy.

6. Bank Onboarding and Policy Configuration

6.1 Bank institution data

ZM-IAM-001 A bank organization record **MUST** capture: legal name, licence number, contact details, signed platform agreement reference, authorized administrators, disbursement and collection account details, and integration configuration.

6.2 Bank states

INVITED → ONBOARDING → UNDER_REVIEW → ACTIVE → SUSPENDED → TERMINATED

ZM-IAM-002 Only an ACTIVE bank may be evaluated for listing eligibility or submit offers.

6.3 Bank policy filters

ZM-MKT-001 Each bank **MUST** be able to configure policy filters that determine which listings it is eligible to see:

Filter	Type
Sector / activity	Include / exclude list
Minimum and maximum invoice amount	Range
Tenor (days to maturity)	Range
Accepted transactionType values	Set
Recourse appetite	Set of recourseType
Minimum composite Trust Score	Threshold
Maximum risk band	Enum ceiling
Required document types	Set
Buyer restrictions	Include / exclude list
Supplier restrictions	Include / exclude list
Geography / governorate	Include / exclude list

ZM-MKT-002 Eligibility **MUST** require **both** conditions: (a) the bank is active and permitted by platform administration, **and** (b) the listing matches the bank's configured filters. ZM-MKT-003 Every eligibility evaluation **MUST** be persisted with its outcome and the specific rules applied, so a bank can be shown why it did or did not receive a listing.

7. Buyer Directory and Resolution

7.1 Buyer as a global entity

ZM-BUY-006 Buyer **MUST** be a **global, platform-wide entity**, uniquely keyed by national establishment number. ZM-BUY-007 Multiple suppliers and multiple invoices **MUST** be able to reference the same Buyer record. ZM-BUY-008 Supplier-specific contact and relationship data **MUST** live on SupplierBuyerRelationship, never on Buyer.

7.2 Data requested from the supplier

Field	Requirement	Use
Buyer establishment name	Required	Registry search
Buyer contact number	Required	Future communication
Contact-person name	Required	Identify the number's owner
Contact-person role	Required	Accountant, purchasing, manager, owner, other

Field	Requirement	Use
Contact email	Optional	Notification channel
Receiving branch	When applicable	Link invoice to correct location

7.3 Resolution workflow

1. Search buyers already linked to this supplier.
2. If none, search buyers already known to the platform.
3. If still none, query CCD by name via the government adapter.
4. Display candidate results: name, type, national number, status, governorate.
5. **The supplier selects the exact buyer** and confirms it is the entity named on the invoice.
6. Platform matches national number, name, and available invoice data.
7. Retrieve the full company profile for the confirmed national number.
8. Create or link the unified Buyer record.
9. Create or update the SupplierBuyerRelationship with contact data.

ZM-BUY-009 The platform **MUST NOT** automatically select a buyer based on name similarity alone, under any circumstances. ZM-BUY-010 Ambiguous, multiple, or unclear results **MUST** move to BUYER_MANUAL_REVIEW.

7.4 Buyer verification states and invoice effect

Buyer state	Effect on invoice
ACTIVE + MATCHED	Proceed
ACTIVE + PARTIAL_MATCH	Manual review or clarification
NOT_FOUND	Manual review; alternative lookup key
NAME_MISMATCH / ID_MISMATCH	Blocked until resolved
SUSPENDED	Blocked
STRUCK_OFF	Blocked
UNDER_LIQUIDATION	Manual review — see LT-02

7.5 Buyer contact data

ZM-BUY-011 Contact data is stored as **supplier-provided**, explicitly not as the buyer's official registry contact. ZM-BUY-012 Contact states: SUPPLIER_PROVIDED, UNVERIFIED, CONTACTED, VERIFIED_BY_CONTACT, INVALID, DO_NOT_CONTACT. ZM-BUY-013 The originating supplier and every subsequent update **MUST** be recorded with actor and timestamp. ZM-BUY-014 DO_NOT_CONTACT **MUST** suppress all outbound notification to that contact, across all suppliers, permanently until explicitly reversed by an administrator with a recorded reason. ZM-BUY-015 The platform **MUST** collect only contact data necessary for the transaction, restrict access by role, protect sensitive fields at rest, audit every access and update, and support correction and invalidation.

8. Invoice Submission

8.1 Required invoice data

Group	Fields
Identification	Invoice number; electronic-invoice identifier; issue date
Parties	Registered supplier; resolved buyer; branch when applicable
Value	Subtotal; tax; total (faceValue); currency; paidAmount; outstandingAmount
Maturity	Due date; payment terms; payment period
Trade documents	Goods/services description; purchase-order number; delivery-note number when available
Financing	minimumAcceptableAmount; disclosure of any related previous financing

ZM-INV-001 `outstandingAmount` = `faceValue` – `paidAmount`, and **MUST** be greater than zero for an invoice to be listed. ZM-INV-002 Currency **MUST** be stored as an ISO 4217 code; the only accepted value in V3 is **JOD**. Invoice, offer, settlement, commission, and payout currencies **MUST** all match. ZM-INV-003 All monetary values **MUST** be stored as `numeric(18,3)` and computed with decimal arithmetic. Floating-point arithmetic on money is prohibited.

8.2 Documents

ZM-DOC-001 An **electronic invoice is mandatory** for every submission in V3. A submission without one cannot proceed. ZM-DOC-002 Delivery note, purchase order, statement of account, and any credit note, return, or adjustment are **mandatory or conditional** per bank or platform policy (CONFIGURABLE). ZM-DOC-003 Every document **MUST** store a content hash, MIME type, size, uploader, and upload timestamp. ZM-DOC-004 Documents **MUST** be held in private storage; access **MUST** be by short-lived signed URL issued only after a server-side authorization check.

8.3 OCR and QR extraction

ZM-DOC-005 OCR **MUST** perform **both** functions: (a) **pre-fill** structured invoice fields for supplier review and correction; (b) **cross-validate** extracted values against supplier-confirmed values and raise mismatches. ZM-DOC-006 Both the original OCR output **and** the supplier's corrections **MUST** be preserved and independently retrievable. A supplier correction **MUST NOT** erase the machine-extracted value. ZM-DOC-007 The platform **MUST** decode and parse the QR code on the Jordanian electronic invoice locally. ZM-DOC-008 QR-derived values **MUST** be compared against OCR output and supplier-entered fields; each comparison result **MUST** be recorded as a `VerificationCheck`. ZM-DOC-009 The platform **MUST** call a **dummy**

government e-invoice validation adapter, replaceable later by a real validation service without core changes. ZM-DOC-010 The system **MUST NOT** hard-code an undocumented official QR payload structure. Parsing **MUST** be schema-driven and tolerant of format variation, degrading to “unparsed — manual review” rather than failing loudly or guessing.

8.4 Supplier declarations

ZM-INV-004 For every invoice the supplier **MUST** affirm, with the declaration text version recorded:

- The invoice is authentic and represents a real commercial transaction.
- The goods or services were delivered per the commercial relationship.
- The invoice is unpaid and not cancelled.
- No known dispute, return, or undisclosed discount exists.
- The invoice has not previously been sold, assigned, pledged, or financed.
- The selected buyer is the entity named on the invoice.
- The contact details belong to a buyer representative the supplier actually uses.
- The supplier accepts recourse and indemnification for false declarations, as set out in the contract.

LEGAL_TBD_POST_COMPETITION — **LT-04**. Enforceability and precise wording of the supplier declarations and indemnity. **Technical assumption:** declaration text is a versioned template; the accepted version is stored per submission.

8.5 Automated checks

Check	Detail	Outcomes
Completeness	Required fields, documents, dates, values	PASS / MISSING
Identity match	Supplier and buyer vs. invoice content	MATCH / MISMATCH
Duplicate detection	Fingerprint over parties, invoice number, date, value, tax	UNIQUE / DUPLICATE
Transaction logic	Maturity sanity, amount, currency, negatives, modifications	PASS / REVIEW
Party eligibility	Supplier and buyer status; exposure limits	ELIGIBLE / BLOCKED
File integrity	Hash, MIME type, visible modification indicators	PASS / FRAUD_REVIEW
OCR consistency	Extracted vs. confirmed values	PASS / MISMATCH
QR consistency	QR vs. OCR vs. confirmed values	PASS / MISMATCH / UNPARSED

ZM-VER-001 The active-invoice **fingerprint MUST be unique** platform-wide. A collision **MUST** block submission and open a review record. ZM-VER-002 A failed check **MUST NOT** by itself be treated as proven fraud. It routes to review.

8.6 Pre-market invoice states

DRAFT → SUBMITTED → AUTOMATED_CHECKS → UNDER_REVIEW →
INFORMATION_REQUIRED → ELIGIBLE | REJECTED | FRAUD_REVIEW

ZM-INV-005 ELIGIBLE means verification passed and the invoice may be listed. It does not imply any financing commitment.

9. Risk, Trust Score, and Machine Learning

9.1 Purpose and limits

ZM-RSK-001 The Trust Score is **decision support only**. It **MUST NOT** be presented as a guarantee, a credit rating, or a substitute for the bank's own credit decision. ZM-RSK-002 Every score display **MUST** carry a plain-language disclaimer to this effect, in both languages.

9.2 Structure

ZM-RSK-003 The platform **MUST** produce both:

- a **composite Trust Score**, numeric **0–100**; and
- a **risk band**: LOW / MEDIUM / HIGH / CRITICAL.

ZM-RSK-004 The platform **MUST** also produce five component indicators:

Indicator	Measures	Example inputs
Supplier Verification	Completeness and reliability of the supplier profile	Registry status, licence, tax, signatory match, bank account verification
Data Confidence	Strength and freshness of underlying sources	Government vs. verified self-declared vs. unverified vs. conflicting; snapshot age
Buyer Profile	Available information about the debtor	Status, company age, activity, capital, platform payment history
Invoice Score	Invoice integrity and eligibility	Completeness, matching, uniqueness, evidence quality, tenor
Platform Behavior	Supplier conduct on the platform	Prior invoices, disputes, duplicates, recourse history, compliance events

9.3 The data-availability rule

ZM-RSK-005 **Government service downtime, an unavailable source, or a field the source does not publish MUST NEVER reduce the Trust Score.** ZM-RSK-006 Such conditions **MUST** instead reduce a separate dataAvailability measure, and where the field is essential, raise an InformationRequest. ZM-RSK-007 What **does** materially affect the score or trigger rejection: refusal to supply essential information, internal contradictions in supplied data, forged or manipulated evidence, and confirmed adverse registry status. ZM-RSK-008 The distinction between “*the source reported adverse information*” and “*the source did not respond*” **MUST** be preserved structurally at every layer, from adapter to score to UI.

9.4 Versioning

ZM-RSK-009 Scoring weights and rules **MUST** be administrator-configurable, but changes **MUST** create a **new RiskModelVersion**, never edit an active version in place. ZM-RSK-010 Every calculated score **MUST** permanently retain the version identifier used, and historical scores **MUST NOT** change when a new version is activated. ZM-RSK-011 Activation of a new version **MUST** be an auditable administrative action with actor, timestamp, and rationale.

9.5 Bank-facing disclosure

ZM-RSK-012 An eligible bank **MUST** see: composite score; the five component scores; risk band; data-confidence level; positive factors; risk factors; structured reason codes; score version; calculation date. ZM-RSK-013 Banks **MUST NOT** receive raw model internals, feature weights, proprietary implementation details, or any hidden sensitive field.

9.6 Machine learning requirements

ZM-RSK-014 V3 **MUST** include a **real, executable ML training and inference pipeline** — not a stub. ZM-RSK-015 The architecture **MUST** separate two concerns:

- **Deterministic rules** govern hard eligibility and hard fraud blockers. An ML model **MUST NOT** be able to override a deterministic blocker.
- **A trained ML model** produces risk and anomaly estimation contributing to the score.

ZM-RSK-016 Where real historical data is unavailable, the pipeline **MUST** use clearly labelled synthetic or demonstration training data, and the limitation **MUST** be disclosed in the UI and in reports. ZM-RSK-017 The pipeline **MUST** provide: model versioning; recorded training metrics; per-prediction explainability (contributing features and direction); and a graceful fallback to rules-only scoring when the model service is unavailable, with the degraded mode visibly flagged. ZM-RSK-018 Document-forgery detection **MAY** begin with deterministic integrity checks and anomaly features. It **MUST NOT** be presented, labelled, or described as production-grade forensic detection.

10. Marketplace and Listings

10.1 Listing creation

ZM-MKT-004 Only an ELIGIBLE invoice may be listed. ZM-MKT-005 At listing activation the platform **MUST**: evaluate eligibility for every active bank; notify eligible banks; open the offer window; and create the **listing-fee obligation** (\$12.2). ZM-MKT-006 The invoice enters OPEN_FOR_OFFERS.

10.2 Deadlines

ZM-MKT-007 Two deadlines govern a listing:

Deadline	Default (CONFIGURABLE)	Measured from
offerSubmissionDeadline	24 hours	Listing activation
supplierSelectionDeadline	12 hours	Close of offer submission

ZM-MKT-008 Defaults are set by platform administration. **The supplier MUST NOT choose arbitrary deadlines in V3.** ZM-MKT-009 The offer window **MUST** close automatically at offerSubmissionDeadline. No offer, revision, or withdrawal is accepted afterwards. ZM-MKT-010 The platform **MUST** send reminders before the supplier-selection deadline (CONFIGURABLE; ASSUMPTION — default at 50% and 15% of remaining time).

10.3 Confidentiality

ZM-MKT-011 A bank **MUST NOT** be able to see, infer, or derive: another bank's identity as a participant; another bank's offered amount; another bank's fees; another bank's conditions; or the total number of competing offers. ZM-MKT-012 The supplier's minimumAcceptableAmount **MUST NOT** be disclosed to any bank, in any form, including through validation error messages or UI hints. ZM-MKT-013 Only the supplier may compare all active offers.

Design note. ZM-MKT-012 has a concrete implementation consequence: when a bank submits an offer whose net payout falls below the supplier's floor, the rejection message **MUST NOT** reveal the floor or the shortfall. It states only that the offer does not currently meet the supplier's requirements.

10.4 Bank-visible data

ZM-MKT-014 An eligible bank **MUST** see the **verified legal identity** of both supplier and buyer, because it cannot underwrite otherwise: company identity, registration details, status, permitted invoice data, permitted documents, Trust Score and indicators, and policy-permitted transaction history. ZM-MKT-015 Personal data **MUST** be minimized and role-restricted. Banks see only the personal, signatory, and contact data necessary for underwriting and contracting.

LEGAL_TBD_POST_COMPETITION — **LT-05.** Lawful basis and consent scope for disclosing supplier, signatory, and buyer personal data to banks. **Technical**

assumption: disclosure gated by a versioned consent record and a configurable field-level disclosure policy per recipient role.

10.5 Listing outcomes

Outcome	Result
Supplier accepts an offer	Invoice locked; proceeds to contracting
Supplier rejects all offers	Invoice returns to ELIGIBLE
Selection deadline lapses	Active offers expire; listing closes; invoice returns to ELIGIBLE
No offers received	Listing closes; invoice returns to ELIGIBLE

ZM-MKT-016 **Relisting is never automatic.** It requires a manual request and platform verification that: the invoice is still unpaid; has not already been financed; has not been cancelled or materially changed; has not expired; and both supplier and buyer remain eligible. ZM-MKT-017 Each relisting round **MAY** incur a further listing fee per CONFIGURABLE policy.

11. Bank Offers

11.1 Offer content

ZM-0FR-001 A BankOffer **MUST** contain:

Classification

Field	Values
transactionType	INVOICE_FINANCING, RECEIVABLE_PURCHASE, RECEIVABLE_ASSIGNMENT, OTHER
recourseType	FULL_RECOURSE, LIMITED_RECOURSE, NON_RECOURSE, OTHER

Money

Field	Meaning
grossFundingAmount	Total the bank commits against the receivable
bankDiscountAmount	Bank's discount
bankFeesAmount	Bank's fees and charges
platformCommissionAmount	Zimmamless success commission
listingFeeAmount	Listing fee applied to this transaction (unpaid portion)
otherDeductionsAmount	Any other agreed deduction
netSupplierPayout	Computed result

Terms

Field	Meaning
expectedPayoutDate	Indicative funding date
validUntil	Offer expiry
conditions[]	Structured OfferCondition records
versionNumber, previousOfferId	Revision lineage

11.2 The money formula

```
netSupplierPayout =  
    grossFundingAmount  
    - bankDiscountAmount  
    - bankFeesAmount  
    - platformCommissionAmount  
    - unpaidListingFeeAmount  
    - otherDeductionsAmount
```

ZM-0FR-002 The supplier **MUST** be shown both the **full component breakdown** and the **single net figure**, side by side, before acceptance. ZM-0FR-003 The formula **MUST** be evaluated server-side with decimal arithmetic. Client-side computation is presentational only and **MUST** be re-verified on submission.

11.3 Amount constraints

ZM-0FR-004 minimumAcceptableAmount is the **minimum net amount the supplier is willing to receive** — not a gross floor.

```
minimumAcceptableAmount > 0  
minimumAcceptableAmount ≤ invoice.outstandingAmount  
grossFundingAmount ≤ invoice.outstandingAmount  
netSupplierPayout ≥ minimumAcceptableAmount
```

ZM-0FR-005 Every constraint **MUST** be enforced server-side at offer submission, at offer revision, and again at acceptance. ZM-0FR-006 **Partial funding is permitted**. A bank may offer less than the full outstanding amount, provided the resulting net payout meets the supplier's floor. ZM-0FR-007 Once an offer is accepted, the invoice is locked in full. The remaining unfunded portion **MUST NOT** be separately listed in V3.

11.4 Transaction type behaviour

ZM-0FR-008 All transaction types **MUST** share the same core lifecycle and marketplace workflow. transactionType **MUST NOT** create a separate state machine in V3. ZM-0FR-009 transactionType **MAY** select: a different contract template; different disclosure text; different required documents; different permitted bank conditions; a different default recourse configuration. ZM-0FR-010 transactionType is set **per offer**. A bank **MAY** define a default in its policy and **MAY** override it per offer. ZM-0FR-011 A supplier **MAY** legitimately be comparing offers of different transaction types simultaneously. The type and its plain-language explanation **MUST** be prominently displayed for each offer before selection. ZM-0FR-012 The

supplier **MUST** see, for every offer, before accepting: transaction type; recourse type; gross amount; every deduction; net payout; all conditions; expected payout date; and offer validity.

LEGAL_TBD_POST_COMPETITION — **LT-06**. The legal effect, contractual meaning, and enforceability of each transactionType, and of each recourseType.

Technical assumption: each is an enum driving a template selection and a disclosure text block; no legal consequence is inferred by the system.

11.5 Offer lifecycle

DRAFT → PENDING_INTERNAL_APPROVAL → ACTIVE → (REVISED)* → SELECTED | NOT_SELECTED | WITHDRAWN | EXPIRED

ZM-0FR-013 A bank has **exactly one current offer** per listing. Revisions supersede; every prior version is retained immutably. ZM-0FR-014 Offers **MUST** be submitted within the open window; submission outside it **MUST** be rejected with a clear reason. ZM-0FR-015 A bank **MAY withdraw an active offer before acceptance with no penalty**. The withdrawal is audited. The penalty regime (\$16) applies **only after** acceptance. ZM-0FR-016

Maker/approver separation applies: an offer reaches ACTIVE only after approval by a different user holding the Offer Approver role.

11.6 Offer conditions

ZM-0FR-017 OfferCondition types: REQUIRED_GUARANTEE, REQUIRED_DOCUMENT, RECOURSE_TERM, FUNDING_TIMELINE, CONTRACTUAL_CONDITION, OTHER. ZM-0FR-018 Each condition **MUST** carry: type, title, description, mandatory flag, display order, and a fulfilment status tracked through to contracting. ZM-0FR-019 REQUIRED_POST_DATED_CHEQUE is **removed** from the condition vocabulary in V3.

12. Offer Selection, Locking, and Fees

12.1 Selection and atomic locking

ZM-SEL-001 The supplier **MAY accept an active offer at any time while the listing is open**. There is no requirement to wait for the submission deadline. ZM-SEL-002 Acceptance **MUST** execute atomically within a single database transaction that:

1. re-validates the offer is still ACTIVE and within validUntil;
2. re-validates netSupplierPayout ≥ minimumAcceptableAmount;
3. re-validates the invoice is still lockable and unmodified;
4. sets the invoice/transaction to locked;
5. marks the selected offer SELECTED;
6. marks every other active offer NOT_SELECTED;
7. creates the immutable AcceptedOfferSnapshot;
8. writes the audit entries.

ZM-SEL-003 If any step fails, the entire transaction **MUST** roll back with no partial state. ZM-SEL-004 A second acceptance **MUST** be impossible. The lock **MUST** be enforced at the database level, not only in application logic. ZM-SEL-005 The platform **MUST NEVER** select an offer automatically, and **MUST NEVER** favour the highest amount. Selection is an explicit supplier act recording `selectedBy` and `selectedAt`. ZM-SEL-006 The selected offer is **not necessarily the highest** — the supplier legitimately weighs conditions, recourse type, transaction type, and timing against the amount.

12.2 The accepted offer snapshot

ZM-SEL-007 `AcceptedOfferSnapshot` **MUST** freeze, immutably: bank identity; supplier identity; invoice reference; transaction type; recourse type; every money component; net payout; all accepted conditions; source offer version; selection timestamp; and a content hash. ZM-SEL-008 It **MUST** remain unchanged even if the source `BankOffer` record is later modified or superseded.

12.3 Listing fee

ZM-FEE-001 The listing fee is **incurred when the marketplace listing becomes active**, regardless of whether financing ultimately succeeds. ZM-FEE-002 At listing activation the platform **MUST** create a **listing-fee obligation** record against the supplier. ZM-FEE-003 The listing fee is **separate and independent** from the success commission. ZM-FEE-004 If unpaid at payout time, the outstanding listing fee **MAY** be deducted from the supplier's net payout, appearing as `unpaidListingFeeAmount` in the offer breakdown. ZM-FEE-005 If no funding occurs, the listing fee **remains recorded as payable**. ZM-FEE-006 Fee amount, calculation basis, and relisting-fee policy **MUST** be CONFIGURABLE. ZM-FEE-007 The supplier **MUST** be shown the listing fee and its terms **before** confirming listing activation.

LEGAL_TBD_POST_COMPETITION — **LT-07**. The legal mechanism for collecting, invoicing, and enforcing the listing fee, particularly where no funding occurs.

Technical assumption: the obligation is recorded in the ledger as a receivable from the supplier; collection mechanics are adapter-driven and configurable.

12.4 Platform commission

ZM-FEE-008 The commission fee-payer **MUST** be CONFIGURABLE: SUPPLIER, BANK, SPLIT, CUSTOM. ZM-FEE-009 The **V3 competition default is SUPPLIER** — the commission is deducted from the supplier's proceeds. ZM-FEE-010 The data model **MUST** support a separate bank-side fee even where the default is supplier-paid. ZM-FEE-011 Commission is calculated from `grossFundingAmount` using the `CommissionTier` active at calculation time. It **MUST NOT** be derived from `faceValue` or `minimumAcceptableAmount`. ZM-FEE-012 `CommissionCalculation` **MUST** store the applied percentage and applied fixed amount as a snapshot, so later tier changes never retroactively alter a completed calculation.

12.5 Commission recognition

ZM-FEE-013 Commission becomes **final only on SUPPLIER_PAYOUT_COMPLETED**. ZM-FEE-014 It is **NOT** finalized when: the offer is accepted; the contract is signed; or the bank merely

initiates funding. ZM-FEE-015 A finalized commission **MUST NEVER** be deleted or edited. Any reversal **MUST** be a **compensating entry** referencing the original.

13. Contracts and Signatures

13.1 Contract generation

ZM-CON-001 The platform **MUST** generate the contract from a **template engine with structured merge fields**, populated from the `AcceptedOfferSnapshot` and verified party data. ZM-CON-002 A **template per transactionType** **MUST** be maintained, with a configurable default fallback template. ZM-CON-003 Templates **MUST** be versioned; the version used **MUST** be recorded on the contract. ZM-CON-004 The bank **MAY** attach additional documents or condition annexes. ZM-CON-005 A `ContractTermSnapshot` **MUST** freeze all terms at generation, with a content hash.

13.2 Pre-contract checks

ZM-CON-006 Before contract generation the platform **MUST** confirm: the invoice has not changed, expired, or been cancelled; mandatory offer conditions are fulfilled or explicitly waived with a record; supplier declarations are reconfirmed; and the supplier bank account is verified.

13.3 Signature

ZM-CON-007 Signature **MUST** operate through a **provider-agnostic SignatureProvider adapter**. ZM-CON-008 For the competition the platform **MUST** ship a dummy/sandbox provider supporting **in-platform click-to-accept**, recording: signer identity, organization, capacity, timestamp, IP address, device metadata, signed-document hash, and a complete audit trail. ZM-CON-009 A production PKI or qualified signature provider **MUST** be insertable later without core domain changes. ZM-CON-010 Default signature requirement: **one authorized supplier signatory and one authorized bank signatory**. The data model **MUST** support multiple required signatories per organization. ZM-CON-011 A signature counts only after `SignatureVerification` confirms document integrity, signer identity, and signer authority. ZM-CON-012 The contract becomes `FULLY_SIGNED` only when every required verified signature is present.

13.4 Zimmamless as a party

ZM-CON-013 Zimmamless **does not sign** the bank–supplier receivable contract in the V3 technical model. ZM-CON-014 Platform fees and usage terms are accepted through a **separate platform agreement / terms acceptance**, versioned and recorded per organization.

LEGAL_TBD_POST_COMPETITION — **LT-08**. Contractual wording of every template; enforceability of click-to-accept signature; whether Zimmamless must be a contract party to charge its fee; and whether any registration, assignment, or perfection step is legally required. **Technical assumption**: two-party contract; separate platform

terms; the data model retains an AssignmentRecord placeholder for future perfection evidence.

13.5 Prior rights and perfection

ZM-CON-015 Zimmamless **does not perform** prior-right checks or legal perfection. These remain entirely with the bank. ZM-CON-016 The platform **MAY** record bank-supplied evidence of any perfection step for completeness.

LEGAL_TBD_POST_COMPETITION — **LT-09**. Whether legal perfection or registration of assignment is required in Jordan, and who bears responsibility. **Technical assumption:** bank responsibility; platform records evidence only.

13.6 Double-financing control

ZM-CON-017 An invoice in OFFER_ACCEPTED, CONTRACTED, FUNDING_CONFIRMATION_PENDING, or FUNDED **MUST NOT** be re-uploaded or re-listed. ZM-CON-018 This is a **platform-internal control only**. It does not replace the bank's own prior-rights checks or any legal perfection outside Zimmamless, and the UI **MUST** say so plainly to banks. —

14. Funding, Cross-Party OTP, and Settlement

14.1 What the OTP is for

The V3 OTP is **not** related to the removed cheque handover. It serves a different purpose entirely.

ZM-FND-001 The funding OTP confirms **two things simultaneously**: (a) the **bank** has marked the funding transfer as executed; and (b) the **supplier** acknowledges the funding confirmation and participates in the final platform confirmation.

ZM-FND-002 Its function is to **prevent the bank from unilaterally moving the transaction to FUNDED**. A bank-only confirmation would be one-sided; the OTP creates evidence that both parties participated.

14.2 Funding sequence

1. Contract is FULLY_SIGNED; all mandatory conditions are fulfilled.
2. Supplier bank account is verified; CommissionCalculation is prepared.
3. Bank executes or confirms the funding transfer.
4. Authorized bank user clicks **Funding Sent** in the Zimmamless bank portal.
5. Bank user **generates the OTP** in the portal.
6. Bank communicates the OTP to the supplier through their agreed channel, at a coordinated time.
7. Supplier enters the OTP into Zimmamless.
8. Platform validates the OTP.
9. Platform checks settlement/payment evidence.

10. **Both** OTP confirmation **and** settlement evidence being satisfied, the transaction becomes FUNDED.

ZM-FND-003 **Both** conditions are required. OTP alone does not produce FUNDED; settlement evidence alone does not produce FUNDED.

14.3 OTP rules

ZM-FND-004 The funding OTP is **mandatory for every V3 transaction**. The architecture **MUST** keep it configurable for future policy change. ZM-FND-005 The OTP **MUST** be: single-use; time-limited; stored only as a secure hash; bound to one specific transaction; bound to the issuing bank user; and fully audited. ZM-FND-006 Defaults (CONFIGURABLE):

Parameter	Default
Validity	15 minutes
Maximum verification attempts	5
Maximum resends	3
Single use	Yes
Storage	Hash only

ZM-FND-007 Every OTP creation, resend, failed attempt, successful validation, and expiry **MUST** be audited with actor, timestamp, and IP. ZM-FND-008 The OTP is explicitly **not** a digital signature and carries **no** legal signing authority. Its sole purpose is cross-party synchronization of the funding confirmation event. ZM-FND-009 Failed attempts **MUST** be rate-limited and **MUST NOT** reveal whether the entered code was close, wrong, expired, or already used beyond a single generic failure message plus remaining-attempt count.

14.4 Pending confirmation

ZM-FND-010 Between “Funding Sent” and successful OTP validation, the transaction sits in **FUNDING_CONFIRMATION_PENDING**. ZM-FND-011 While pending, the platform **MUST**: send reminders to the supplier; permit OTP regeneration by the bank; **escalate to platform administration after 24 hours** (CONFIGURABLE); **NOT** mark the invoice FUNDED; and **NOT** finalize the commission. ZM-FND-012 The transaction **MUST NOT** stall silently. Escalation creates an administrative task with full context.

14.5 Settlement architecture

ZM-FND-013 The technical settlement model is:

Bank

- Zimamless settlement workflow (via configurable SettlementProvider adapter)
- platform commission calculated and withheld
- unpaid listing fee withheld
- net supplier payout

ZM-FND-014 The system **MUST** record three distinct financial legs: **gross bank funding**, **Zimamless deductions** (commission + listing fee + other), and **net supplier payout**. ZM-

FND-015 The platform **MUST** use a **provider-neutral SettlementProvider adapter** with dummy and production implementations. No provider name appears in the core domain model.

ZM-FND-016 The architecture **MUST NOT** assume Zimmamless may hold customer funds in an ordinary operating account.

LEGAL_TBD_POST_COMPETITION — **LT-10**. Whether routing funds through the platform requires a Central Bank of Jordan payment-service licence; whether funds may pass through a controlled account or must use a licensed provider performing split settlement. **Technical assumption:** an abstract settlement adapter with explicit split-settlement support, so the final licensed structure can be inserted without redesign.

14.6 Payout failure

ZM-FND-017 If the bank funds the settlement but the supplier payout fails, the settlement enters **PAYOUT_FAILED** and the platform **MUST**:

- retry automatically a limited, configurable number of times with backoff;
- escalate to manual review if retries are exhausted;
- **NOT** mark the invoice FUNDED;
- **NOT** finalize the commission;
- guarantee **no duplicate payout** can occur under any retry or concurrency condition;
- audit every attempt with provider reference and failure reason.

ZM-FND-018 Idempotency **MUST** be enforced on every settlement instruction using a stable idempotency key, so a retried or duplicated instruction cannot move money twice. ZM-FND-019 If a completed commission later requires reversal, the system **MUST** create a **compensating reversal entry**, never delete or edit the original record.

14.7 Ledger

ZM-FEE-016 The platform **MUST** operate a **double-entry ledger** for all platform-controlled financial legs: listing fees; platform commissions; gross funding settlement; supplier payout; reversals; and recourse settlement where routed through the platform. ZM-FEE-017 Every ledger entry **MUST** be immutable and append-only. Corrections are made by compensating entries. ZM-FEE-018 Operational records of **bank-reported buyer collection** **MUST** be linked to reconciliation records and **MUST NOT** imply that Zimmamless ever held those funds. The ledger must make this distinction structurally obvious. ZM-FEE-019 The ledger **MUST** balance at all times; an out-of-balance condition is a critical alert.

15. Post-Funding Tracking

15.1 Continued involvement

ZM-PMT-001 Zimmamless **MUST** continue tracking the receivable after funding. Platform involvement does **not** end at disbursement.

The lifecycle continues:

FUNDED

- PARTIALLY_PAID
- PAID
- OVERDUE_UNCONFIRMED
- OVERDUE
- RECOURSE / DISPUTE
- CLOSED

ZM-PMT-002 Rationale, all of which are product requirements: monitor the receivable lifecycle; maintain accurate outstanding balances; improve future Trust Scores with real payment history; build platform-wide buyer payment-behaviour data; and support the overdue, recourse, dispute, and fraud workflows.

15.2 Buyer collection is external

ZM-PMT-003 At maturity the buyer pays the **selected bank directly**. ZM-PMT-004 Zimmamless **MUST NOT**: execute the payment; hold the money; route the money; or guarantee collection. ZM-PMT-005 The bank reports payment through an API or authorized manual entry; Zimmamless records and reconciles the reported status.

15.3 Maturity monitoring

ZM-PMT-006 The platform **MUST** send maturity notifications at **30, 14, and 7 days** before the due date, on the due date, and thereafter per policy (CONFIGURABLE). ZM-PMT-007 A scheduled job **MUST** monitor maturity dates continuously.

15.4 The unconfirmed-overdue distinction

ZM-PMT-008 If the due date passes with no full payment reported, the transaction **MUST** become **OVERDUE_UNCONFIRMED** — not OVERDUE. ZM-PMT-009 The bank **MUST** be notified and asked to confirm the actual payment status. ZM-PMT-010 After bank confirmation the transaction resolves to PAID, PARTIALLY_PAID, or OVERDUE. ZM-PMT-011 **The absence of a bank report MUST NEVER be treated as proven default**. This distinction must be preserved in the data model, the UI, and every report and score input.

15.5 Recording payment

ZM-PMT-012 An authorized bank user records: payment amount, date, provider or bank reference, and evidence. ZM-PMT-013 The platform reconciles cumulative payments against the receivable and recalculates the outstanding balance. ZM-PMT-014 Full payment → PAID, then CLOSED with `closureReason = PAID_IN_FULL`. ZM-PMT-015 Partial payment → PARTIALLY_PAID with a recalculated balance. ZM-PMT-016 Maturity passed with an outstanding balance and bank confirmation → OVERDUE, with an overdue-day counter.

15.6 Supplier visibility

ZM-PMT-017 The supplier **MUST** be able to see: payment status; paid amount; outstanding amount; due date; overdue days; and recourse or dispute status. ZM-PMT-018 The supplier

MUST NOT see: confidential bank reconciliation evidence; internal bank notes; the bank's own credit assessment; or any other bank's data.

15.7 Terminal state

ZM-PMT-019 The single terminal state is **CLOSED**, with a mandatory closureReason:

closureReason	Meaning
PAID_IN_FULL	Buyer paid the bank in full
RECOURSE_SETTLED	Supplier settled under recourse
WRITTEN_OFF	Bank wrote off the balance
DEFAULTED	Confirmed default, unrecovered
CANCELLED_BEFORE_FUNDING	Terminated before any funding
SETTLED_BY_AGREEMENT	Parties reached their own settlement
OTHER	Requires a free-text explanation

ZM-PMT-020 Financial history **MUST** remain immutable after closure. A closed transaction may be viewed and reported on, never edited.

16. Recourse, Disputes, and Bank Withdrawal

16.1 Recourse initiation

ZM-REC-001 **Only an authorized bank user may initiate recourse.** ZM-REC-002 Platform administration **MAY**: review; facilitate; request information; record outcomes; and correct operational errors with audit evidence. ZM-REC-003 Administration **MUST NOT** initiate recourse on the bank's behalf.

16.2 Recourse reasons

Reason code	Example
INVALID_INVOICE	False invoice or materially incorrect data
HIDDEN_DISPUTE_OR_RETURN	Not disclosed before funding
DOUBLE_FINANCING	Previously sold, assigned, pledged, or financed
NON_DELIVERY	Commercial transaction incomplete
NON_PAYMENT	Where payment risk sits with the supplier under the agreed recourse type
OTHER	Requires explanation

16.3 Recourse workflow

RECOURSE_INITIATED → SUPPLIER_NOTIFIED → PAYMENT_PENDING → SETTLED | DISPUTED | LEGAL_ESCALATION

ZM-REC - 004 A recourse request **MUST** record the reason code, requested amount, and supporting evidence. ZM-REC - 005 The supplier **MUST** be notified and **MAY** pay, provide evidence, or dispute the request under the contract. ZM-REC - 006 The platform records the case and evidence. **It does not become a collector or an adjudicator.**

16.4 Recourse settlement

ZM-REC - 007 V3 **MUST** technically support supplier-to-bank recourse repayment through the same configurable settlement-provider architecture. ZM-REC - 008 The system records: requested recourse amount; supplier repayment; provider reference; settlement status; evidence; and remaining balance. ZM-REC - 009 Commission is **not automatically refunded** when recourse occurs. ZM-REC - 010 Any commission refund or adjustment policy **MUST** be CONFIGURABLE, and any adjustment **MUST** be a compensating ledger entry.

LEGAL_TBD_POST_COMPETITION — **LT-11.** Whether commission is refundable on recourse; enforceability of recourse terms; whether platform-routed recourse repayment creates any regulated activity. **Technical assumption:** no automatic refund; configurable policy; compensating entries only.

16.5 Commercial disputes

ZM-REC - 011 The platform opens a Dispute record storing reason, amount, and evidence. ZM-REC - 012 Zimnamless **does not decide** the dispute and **does not bear** its value. ZM-REC - 013 Automated state changes on the affected transaction **MUST** pause while a dispute is open. ZM-REC - 014 The case is referred to the bank and supplier under their contract; the platform records the resolution the parties submit.

16.6 Bank withdrawal after acceptance

ZM-REC - 015 If a bank withdraws after its offer has been accepted, the platform **MUST** create a WithdrawalCase recording:

- withdrawal reason code;
- responsible party;
- supporting evidence;
- penalty applicability;
- configurable penalty amount;
- relisting eligibility;
- administrative decision and rationale.

Reason codes and default penalty treatment

Reason code	Default penalty treatment (CONFIGURABLE)
BANK_COMMERCIAL_DECISION	Penalty may apply
SUPPLIER_MISREPRESENTATION	No bank penalty; possible fraud review of the supplier
FRAUD_DISCOVERED	Normally no penalty against the bank
INVOICE_CHANGED	Manual review

Reason code	Default penalty treatment (CONFIGURABLE)
CONDITION_NOT_MET	Manual review
TECHNICAL_FAILURE	Manual review
OTHER	Manual review

ZM-REC - 016 For the competition, the platform **records and calculates** the penalty per configurable policy.

LEGAL_TBD_POST_COMPETITION — **LT-12.** Whether the withdrawal penalty is legally enforceable, and whether it is contractually invoiced or directly deducted. Also whether the supplier is entitled to compensation. **Technical assumption:** the penalty is calculated and recorded; no automatic deduction is executed; supplier compensation is not implemented.

16.7 Post-withdrawal relisting

ZM-REC - 017 The invoice **MUST NOT** return to the marketplace automatically. It first enters **manual review**. ZM-REC - 018 Before relisting, the platform **MUST** confirm: no funding occurred; the invoice is still unpaid; it has not changed; it remains valid; no fraud indicator exists; and both supplier and buyer remain eligible. ZM-REC - 019 After review it may return to ELIGIBLE and then be manually relisted.

16.8 Cancellation and amendment rules

Stage	Rule
DRAFT	Supplier may edit or delete freely
SUBMITTED / UNDER_REVIEW	Supplier requests withdrawal; the record is preserved
OPEN_FOR_OFFERS	Cancellation permitted per policy; active offers are closed
OFFER_ACCEPTED	Cancellation subject to offer terms and bank approval
CONTRACTED / FUNDED	No unilateral cancellation. Settlement and a documented decision are required

17. Fraud and Compliance

ZM-FRD - 001 On a fraud indicator the platform **MUST**: freeze the invoice; stop funding if it has not yet occurred; open FRAUD_REVIEW; and notify Compliance. ZM-FRD - 002 The platform **MAY** request additional evidence and suspend new invoice submissions by the supplier during review. ZM-FRD - 003 Fraud decision states: CLEARED, RESTRICTED, SUSPENDED, BLACKLISTED, REPORTED. ZM-FRD - 004 A fraud case is a **suspected** case requiring investigation. It **MUST NOT** be presented as confirmed fraud until a Compliance Officer records that determination. ZM-FRD - 005 A failed VerificationCheck or a high risk score **MUST NOT** automatically create a confirmed-fraud outcome. ZM-FRD - 006 Fraud cases **MUST** support structured evidence, case assignment, and a full decision audit trail.

LEGAL_TBD_POST_COMPETITION — **LT-13.** Mandatory reporting obligations, sanctions-screening requirements, AML/CFT duties, and the legal consequences of blacklisting. **Technical assumption:** the REPORTED state and a screening adapter exist; no specific reporting integration is implemented in V3.

18. Notifications

18.1 Channels

ZM-NOT-001 V3 channels: **Email, WhatsApp, in-platform**, and **optional manual phone-call recording**. ZM-NOT-002 **SMS is not required** in V3. The channel abstraction **MUST** permit adding it later without core change.

18.2 Buyer notification

ZM-NOT-003 Zimmamless sends the buyer notification **on behalf of the confirmed transaction**, identifying the selected bank. ZM-NOT-004 It is sent **after** the transaction is confirmed between bank, platform, and supplier. ZM-NOT-005 Content for the competition version is **operational**: confirmation that the receivable transaction is finalized; the selected bank's identity; the invoice reference; relevant payment or communication instructions; and contact details for questions. ZM-NOT-006 The notification **MUST NOT** request or imply any buyer approval, rejection, or decision authority.

LEGAL_TBD_POST_COMPETITION — **LT-14.** Exact legal wording; whether the notification constitutes formal notice of assignment; whether acknowledgement is legally required; and required disclosures. **Technical assumption:** versioned notification templates per transaction type and language; operational wording only; full delivery evidence retained.

18.3 Delivery evidence

ZM-NOT-007 For every notification the platform **MUST** store: channel; destination; send time; provider reference; delivery status; failure reason; retry count; and, where applicable, the manual call record with the recording user and outcome. ZM-NOT-008 Delivery evidence **MUST** be immutable and available in the audit trail.

18.4 Internal notifications

ZM-NOT-009 The platform **MUST** notify at minimum: application state changes; information requests; SLA warnings; new eligible listings (banks); offer received (supplier); offer accepted/not selected (banks); contract ready for signature; funding sent; OTP pending; payout completed or failed; maturity reminders; overdue; recourse; dispute; withdrawal case; and fraud freeze. ZM-NOT-010 Notification templates **MUST** be versioned and bilingual.

19. Audit, Administration, and Reporting

ZM-AUD-001 Every change to invoices, offers, contracts, payments, fees, scores, and decisions **MUST** generate an immutable AuditLog entry. ZM-AUD-002 An audit entry **MUST** record: actor user; actor organization (active context); action type; target entity type and identifier; before and after snapshots; IP address; device information; correlation identifier; and timestamp. ZM-AUD-003 The audit log is **append-only**. No user, including a Super Admin, may edit or delete an entry. ZM-AUD-004 StatusHistory **MUST** preserve every state transition for all significant entities. Domain logic **MUST NOT** depend solely on a current-status column where history matters. ZM-AUD-005 **Financial records and audit logs MUST NEVER be deleted.**

19.1 Uniqueness rules

ZM-AUD-006 National establishment number **MUST** be unique per supplier and per buyer platform-wide. ZM-AUD-007 Active invoice fingerprint **MUST** be unique platform-wide.

19.2 Permission matrix

Operation	Supplier	Bank	Administration
Edit supplier profile	Before submission, or via change request	No	Review and approve
Upload invoice	Yes	No	Documented support only
View offers	Own invoices only	Own offers only	Per permission
Submit offer	No	Yes	No
Approve offer internally	No	Yes (different user)	No
Accept offer	Yes	No	No
Sign contract	Authorized signatory	Authorized signatory	No
Mark funding sent / generate OTP	No	Yes	No
Enter funding OTP	Yes	No	No
Record buyer payment	Read only	Yes	Review
Initiate recourse	No	Yes	No
Open fraud review	Report only	Report only	Yes
Suspend account	No	No	Yes
Delete financial record	No	No	No
View audit log	Limited (own)	Limited (own)	Full, for authorized roles

19.3 Reporting

ZM-AUD-008 The platform **MUST** provide role-scoped reports covering: onboarding funnel and SLA performance; listing and offer activity; conversion; funding volume and value; fee and commission revenue; payment performance and ageing; overdue and recourse; dispute and fraud; and score distribution by version. ZM-AUD-009 Every report **MUST** respect the same confidentiality rules as the UI — no report may leak another bank's offer data or a supplier's private floor.

20. Internationalization

ZM-I18N-001 The platform **MUST** support **English** and **Arabic** with complete translation of all user-facing text, including validation messages, emails, notifications, and generated documents. ZM-I18N-002 **Full RTL layout MUST** be supported for Arabic — not merely text direction, but mirrored layout, iconography, navigation, tables, and form flow. ZM-I18N-003 **English is the default display language** for all users, regardless of browser or device locale. The platform **MUST NOT** auto-select Arabic from locale detection. ZM-I18N-003a Language switching **MUST** be available at any time, from any screen, and the chosen language **MUST** persist per user across sessions and devices. ZM-I18N-003b English is the **canonical language** for contracts, notifications, and generated documents. Where an Arabic version exists, the English version governs in case of discrepancy. LEGAL_TBD_POST_COMPETITION — see LT-17. ZM-I18N-004 Dates, numbers, and currency **MUST** be localized. Currency **MUST** display as JOD with **three decimal places**. ZM-I18N-005 Contract and notification templates **MUST** exist in both languages, versioned independently, with a recorded canonical language per document. ZM-I18N-006 Bidirectional text mixing (Arabic prose containing Latin company names or IBANs) **MUST** render correctly. —

21. Consolidated State Machines

21.1 Supplier application

DRAFT

- SUBMITTED
- AUTOMATED_VERIFICATION
- UNDER_REVIEW
- ⇌ INFORMATION_REQUIRED ⇌ INFORMATION_RESUBMITTED
- ⇌ GOVERNMENT_SERVICE_UNAVAILABLE
- FINAL_REVIEW
- APPROVED | APPROVED_CONDITIONAL | REJECTED

APPROVED → ACTIVE → SUSPENDED | RESTRICTED | TERMINATED

21.2 Bank organization

INVITED → ONBOARDING → UNDER_REVIEW → ACTIVE → SUSPENDED → TERMINATED

21.3 Buyer resolution

SEARCH_INITIATED

- CANDIDATES_PRESENTED
- SUPPLIER_SELECTED

- REGISTRY_LOOKUP
- MATCHED | PARTIAL_MATCH | NOT_FOUND | MISMATCH | BLOCKED
- BUYER_MANUAL_REVIEW (where ambiguous)
- BUYER_RESOLVED

21.4 Invoice / ReceivableTransaction

#	State	Notes
1	DRAFT	Supplier editing
2	SUBMITTED	Awaiting checks
3	AUTOMATED_CHECKS	Machine verification
4	UNDER_REVIEW	Human review
5	INFORMATION_REQUIRED	Awaiting supplier
6	ELIGIBLE	May be listed
7	OPEN_FOR_OFFERS	Listed; offer window open
8	OFFER_ACCEPTED	Locked; atomic
9	CONDITIONS_PENDING	Fulfilling offer conditions
10	CONTRACTED	Fully signed
11	READY_FOR_DISBURSEMENT	All prerequisites met
12	FUNDING_CONFIRMATION_PENDING	Funding sent; awaiting OTP
13	FUNDED	OTP + settlement evidence both confirmed
14	PARTIALLY_PAID	Partial buyer payment reported
15	PAID	Full buyer payment reported
16	OVERDUE_UNCONFIRMED	Past due; no bank report
17	OVERDUE	Past due; bank-confirmed unpaid
18	RECOURSE_ACTIVE	Recourse case open
19	DISPUTED	Dispute open; automation paused
20	FRAUD_REVIEW	Frozen
21	CLOSED	Terminal; requires closureReason
—	REJECTED	Terminal; pre-listing

#	State	Notes
—	CANCELLED	Terminal; pre-funding

21.5 Bank offer

DRAFT

- PENDING_INTERNAL_APPROVAL
- ACTIVE
- ⇒ REVISED
- SELECTED | NOT_SELECTED | WITHDRAWN | EXPIRED

21.6 Settlement

PENDING

- FUNDING_RECEIVED
- PAYOUT_INITIATED
- PAYOUT_COMPLETED (→ commission finalized)
- | PAYOUT_FAILED → RETRYING → MANUAL_REVIEW
- | REVERSED (compensating entry only)

21.7 Funding OTP

NOT_REQUIRED (config) | PENDING_GENERATION

- SENT
- VERIFIED | EXPIRED | FAILED_MAX_ATTEMPTS
- REGENERATED (loops to SENT, up to max resends)

21.8 Recourse

RECOURSE_INITIATED → SUPPLIER_NOTIFIED → PAYMENT_PENDING

→ SETTLED | DISPUTED | LEGAL_ESCALATION

21.9 Withdrawal case

WITHDRAWAL_REQUESTED → UNDER_REVIEW

- PENALTY_ASSESSED | NO_PENALTY
- RELISTING_APPROVED | RELISTING_DENIED
- CLOSED

22. Architecture Direction

Detailed in deliverable 5; recorded here for consistency.

Layer	Technology
Web frontend	Next.js, React, TypeScript — responsive, bilingual, RTL
Backend API	Node.js with NestJS; REST with OpenAPI
Database	Supabase PostgreSQL
Authentication	Supabase Auth
Document storage	Supabase Storage (private buckets)

Layer	Technology
OCR / ML service	Python with FastAPI
Integrations	Adapter pattern with dummy and production implementations

Planned hosting: Vercel (frontend); Render, Railway, Fly.io or equivalent (Node and Python services); Supabase (database, auth, private storage).

ZM-ARC - 001 The platform **MUST** be fully deployable and reachable, even though the competition demonstration may also run locally. ZM-ARC - 002 No native mobile application is in scope. The product is a **responsive web application**.

22.1 Authorization architecture

ZM-ARC - 003 Authorization operates in **two mandatory layers**:

- **NestJS is the primary business authorization and workflow layer.** All core financial and state-changing writes go through it.
- **Supabase Row Level Security is mandatory defense in depth**, not an optional extra.

ZM-ARC - 004 The Supabase **service-role key MUST NEVER** be exposed to the frontend or to any client-side bundle. ZM-ARC - 005 RLS policies **MUST** be tested independently of application logic; an RLS policy that only passes because NestJS filtered the query first is not an acceptable policy.

22.2 Adapters

ZM-ARC - 006 The following **MUST** be behind provider-neutral adapter interfaces, each with a dummy and a production implementation:

Adapter	Purpose
GovernmentRegistryProvider	CCD, ISTD, GAM
EInvoiceValidationProvider	Jordanian e-invoice / QR validation
SettlementProvider	Bank-to-supplier settlement and split payout
SignatureProvider	Electronic signature and verification
NotificationProvider	Email, WhatsApp
ScreeningProvider	Sanctions and adverse-media screening
RiskModelProvider	ML inference with rules-only fallback

ZM-ARC - 007 Replacing any dummy adapter with a production implementation **MUST NOT** require changes to core domain logic, database schema, or API contracts.

23. Non-Functional Requirements

23.1 Security

ZM-NFR-001 All traffic over TLS; HSTS enabled. ZM-NFR-002 Passwords hashed with a modern memory-hard algorithm; MFA available and **required** for all bank and platform-admin roles. ZM-NFR-003 Documents in private storage; access only via short-lived signed URLs issued after a server-side authorization check. ZM-NFR-004 Sensitive fields (IBAN, national ID, contact numbers) encrypted at rest and masked in all UI and logs by default. ZM-NFR-005 OTP stored as hash only; rate-limited; attempt-capped. ZM-NFR-006 Idempotency keys mandatory on every settlement and webhook operation. ZM-NFR-007 Webhook events deduplicated by provider event identifier with a processed flag. ZM-NFR-008 All destructive or financial operations require re-authentication or step-up confirmation. ZM-NFR-009 Every action authorized server-side. Client-side checks are presentational only. ZM-NFR-010 Cross-organization data access is a **critical severity** defect class with dedicated regression tests.

23.2 Data integrity

ZM-NFR-011 Money as numeric(18,3); decimal arithmetic only; floating point prohibited on monetary values. ZM-NFR-012 Offer acceptance, settlement, and ledger writes execute in explicit database transactions with appropriate isolation. ZM-NFR-013 Uniqueness constraints enforced at the database level, not only in application code. ZM-NFR-014 Soft deletion only; no hard delete of financial, audit, or transactional records.

23.3 Performance and availability

ZM-NFR-015 Interactive pages target p95 under 2 seconds under demonstration load. ZM-NFR-016 Government and settlement adapter calls run asynchronously with visible progress, timeouts, retries, and circuit breakers. ZM-NFR-017 Adapter unavailability **MUST** degrade gracefully into a paused, clearly-labelled state — never into an error the user cannot interpret or a silent failure.

23.4 Observability

ZM-NFR-018 Structured logging with correlation identifiers propagated across services. ZM-NFR-019 Metrics on SLA compliance, adapter health, settlement success, OTP completion, and score-service availability. ZM-NFR-020 Alerts on ledger imbalance, payout failure, stalled funding confirmation, and RLS policy violation attempts.

23.5 Accessibility

ZM-NFR-021 Target WCAG 2.1 AA: keyboard navigation, contrast, focus states, screen-reader labelling in both languages.

23.6 Retention

ZM-NFR-022 Retention periods are CONFIGURABLE per data category.

LEGAL_TBD_POST_COMPETITION — **LT-15**. Statutory retention periods for company data, transaction records, documents, and personal data; erasure rights and

their interaction with financial-record immutability. **Technical assumption:** long retention by default; configurable per category; personal-data fields separable from financial records so contact data can be invalidated without touching immutable ledger entries.

LEGAL_TBD_POST_COMPETITION — **LT-16.** Whether Jordan-only data residency is mandatory. **Technical assumption:** hosting region is a deployment configuration; no application logic depends on provider location.

24. Demonstration and Test Data

24.1 Seed data

ZM-DEMO-001 The environment **MUST** seed at minimum: **3 banks, 3 suppliers, 6 buyers, 12 invoices**, multiple bank users across every role, **1 platform admin, 1 supplier reviewer, 1 compliance officer, 1 support agent**.

ZM-DEMO-002 Seeded scenarios **MUST** include:

Scenario	Demonstrates
Successful funding, end to end	The happy path
Multiple competing offers	Confidential comparison and non-highest selection
Information required	SLA pause and resume
Duplicate invoice	Fingerprint collision handling
Fraud review	Freeze and compliance workflow
Failed payout	PAYOUT_FAILED, retry, no duplicate payout
Full payment	PAID → CLOSED
Partial payment	PARTIALLY_PAID with recalculated balance
Overdue	OVERDUE_UNCONFIRMED → bank confirmation → OVERDUE
Recourse	Bank-initiated recourse through to settlement
Bank withdrawal after acceptance	WithdrawalCase, penalty assessment, manual relisting

24.2 Demo time machine

ZM-DEMO-003 A **demo-only time-control feature MUST** exist, able to: advance invoice maturity; trigger reminder notifications; simulate overdue status; demonstrate partial payment; and demonstrate recourse. ZM-DEMO-004 It **MUST** be disabled and inaccessible in production environments, gated by environment configuration **and** a server-side guard — not by hiding the UI. ZM-DEMO-005 Every time-machine action **MUST** be audited and clearly flagged in the record as simulated.

Appendix A — Legal TBD Register

ID	Description	Module	Current technical assumption	Specialist required	Decision needed by
LT-01	Whether buyer notification is formal assignment notice; whether acknowledgment is required; legal basis for processing supplier-provided buyer contact data	BUY, NOT	Operational notification; full delivery evidence; contact data correctable and invalidatable; role-restricted	Commercial + data protection counsel	Before production launch
LT-02	Treatment of companies under liquidation or insolvency (supplier and buyer)	SON, BUY	Configurable policy; default manual review escalating to rejection	Insolvency counsel + credit policy	Before production launch
LT-03	Permissible storage, sharing, reuse, and retention of government-sourced data; consent scope for onward disclosure to banks	GOV	Versioned per-purpose consent; snapshots retained; bank disclosure policy-gated	Data protection counsel + registry liaison	Before real API integration
LT-04	Enforceability and precise wording of supplier	INV, CON	Versioned declaration template; accepted	Commercial counsel	Before production launch

ID	Description	Module	Current technical assumption	Specialist required	Decision needed by
LT-05	declarations and indemnity Lawful basis and consent scope for disclosing supplier, signatory, and buyer personal data to banks	MKT	version stored per submission Versioned consent; field-level disclosure policy per recipient role	Data protection counsel	Before production launch
LT-06	Legal effect and enforceability of each transactionType and each recourseType	OFR, CON	Enums driving template and disclosure selection; no legal consequence inferred by the system	Commercial + banking counsel	Before production launch
LT-07	Legal mechanism for collecting, invoicing, and enforcing the listing fee, especially where no funding occurs	FEE	Recorded as a ledger receivable; collection adapter-driven and configurable	Commercial counsel	Before charging real fees
LT-08	Contract wording; enforceability of click-to-accept signature; whether Zimmamless must be a contract	CON	Two-party contract; separate versioned platform terms; Assignment Record placeholder retained	Commercial counsel + e-signature specialist	Before production launch

ID	Description	Module	Current technical assumption	Specialist required	Decision needed by
	party to charge its fee				
LT-09	Whether legal perfection or registration of assignment is required in Jordan, and who is responsible	CON	Bank responsibility; platform records evidence only	Secured-transactions counsel	Before production launch
LT-10	Whether routing funds through the platform requires a CBJ payment-service licence; controlled account vs. licensed split settlement	FND, FEE	Abstract settlement adapter with explicit split-settlement support	Financial-regulatory counsel	Before any real money movement
LT-11	Commission refundability on recourse; enforceability of recourse terms; regulatory status of platform-routed recourse repayment	REC, FEE	No automatic refund; configurable policy; compensating entries only	Commercial + regulatory counsel	Before production launch
LT-12	Enforceability of the bank withdrawal penalty; invoiced	REC	Calculated and recorded; no automatic deduction;	Commercial counsel	Before production launch

ID	Description	Module	Current technical assumption	Specialist required	Decision needed by
LT-13	vs. deducted; supplier compensation entitlement Mandatory reporting, sanctions screening, AML/CFT duties; legal consequences of blacklisting	FRD	no compensation implemented REPORTED state and screening adapter exist; no reporting integration in V3	AML/ compliance specialist	Before production launch
LT-14	Exact legal wording of buyer notification; formal-notice status; required acknowledgment and disclosures	NOT	Versioned bilingual templates; operational wording; delivery evidence retained	Commercial counsel	Before production launch
LT-15	Statutory retention periods; erasure rights vs. financial-record immutability	AUD	Long default retention; configurable per category; personal data separable from immutable ledger	Data protection counsel	Before production launch
LT-16	Whether Jordan-only data residency is mandatory	ARC	Hosting region is deployment configuration ; no logic depends on provider location	Data protection + regulatory counsel	Before production launch
LT-17	Whether English may	I18N, CON	English canonical	Commercial counsel +	Before production

ID	Description	Module	Current technical assumption	Specialist required	Decision needed by
	serve as the governing language of contracts executed in Jordan, or whether an Arabic version must govern for enforceability or filing		and governing; Arabic provided as an accurate translation; both versions stored and versioned independently per contract	Jordanian litigation counsel	launch

Highest urgency: LT-10 (payment licensing) and LT-13 (AML) are the two items that could materially alter the operating model rather than merely the paperwork. They should be raised first.

Appendix B — Open Assumptions Requiring Owner Confirmation

These were not covered by the answered questions and were filled by the authoring team. Each is safe to override.

ID	Assumption	Where
AS-01	Offer acceptance requires the Supplier Owner/Admin role by default; configurable to allow Invoice Uploader	\$3.3.2
AS-02	Selection-deadline reminders fire at 50% and 15% of remaining time	\$10.2
AS-03	Settlement retry policy: 3 automatic attempts with exponential backoff before manual review	\$14.6
AS-04	Escalation of stalled funding confirmation goes to Operations Admin, not Super Admin	\$14.4
AS-05	Trust Score band thresholds: LOW $\geq$ 75, MEDIUM 50–74, HIGH 25–49, CRITICAL $<$ 25	\$9.2

ID	Assumption	Where
AS-06	Listing fee is a flat configurable amount rather than a percentage in V3	\$12.3
AS-07	Invoices with a due date already in the past are ineligible for listing	\$8
AS-08	Minimum tenor for listing eligibility is 7 days to maturity	\$10
AS-09	A supplier may hold multiple concurrent active listings, subject to a configurable exposure cap	\$10

Confirmed and removed from this register: the former **AS-10** (default display language) was confirmed by the product owner on 22 July 2026 as **English default, Arabic via switcher**, and is now a binding requirement at ZM-I18N-003. All other assumptions above were confirmed as written on the same date.

Appendix C — Glossary

Term	Definition
CCD	Companies Control Department — Jordan's companies registry
ISTD	Income and Sales Tax Department
GAM	Greater Amman Municipality (or equivalent licensing authority)
Buyer / debtor	The company that owes the invoice; never a platform user
Face value	Original total invoice amount
Outstanding amount	faceValue – paidAmount; the amount available to fund
Gross funding amount	Total the bank commits against the receivable
Net supplier payout	What the supplier actually receives after all deductions
Minimum acceptable amount	The supplier's private minimum net payout floor
Listing fee	Charged at listing activation, regardless of outcome
Platform commission	Success fee, finalized only on completed

Term	Definition
Trust Score	payout Composite 0–100 decision-support indicator with five components
Fingerprint	Deterministic hash over invoice identity fields used for duplicate detection
Snapshot	Immutable frozen copy of data at a decision point
Compensating entry	A reversal ledger entry; never a deletion or edit
Adapter	Provider-neutral interface with dummy and production implementations
LEGAL_TBD_POST_COMPETITION	A legal question deferred to post-competition specialist review

Appendix D — Requirements Traceability Summary

Module	Requirement range	Count
IAM / ROL	ZM-IAM-001-002, ZM-ROL-001-008	10
SON	ZM-SON-001-013	13
GOV	ZM-GOV-001-009	9
BUY	ZM-BUY-001-015	15
INV	ZM-INV-001-005	5
DOC	ZM-DOC-001-010	10
VER	ZM-VER-001-002	2
RSK	ZM-RSK-001-018	18
MKT	ZM-MKT-001-017	17
OFR	ZM-OFR-001-019	19
SEL	ZM-SEL-001-008	8
FEE	ZM-FEE-001-019	19
CON	ZM-CON-001-018	18
FND	ZM-FND-001-019	19
PMT	ZM-PMT-001-020	20
REC	ZM-REC-001-019	19
FRD	ZM-FRD-001-006	6
NOT	ZM-NOT-001-010	10
AUD	ZM-AUD-001-009	9
I18N	ZM-I18N-001-006	8

Module	Requirement range	Count
	(incl. 003a, 003b)	
ARC	ZM-ARC-001-007	7
NFR	ZM-NFR-001-022	22
DEMO	ZM-DEMO-001-005	5
Total		288

Approval

This document is deliverable **1 of 6**. On approval, the next deliverables follow in order:

2. Updated Domain / Class Model
3. Updated ERD
4. Updated Workflow and State Diagrams
5. System Architecture
6. Detailed Implementation Plan

No application code will be written until deliverable 6 is approved.

End of Zimmamless V3 Consolidated System Requirements.